Reaction Network Equations

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CHEMICAL REACTION NETWORK
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R1:  ∅  →  X1    (k = 1.0000)
R2:  2X1 + Y1  →  3X1    (k = 1.0000)
R3:  B1 + X1  →  D1 + Y1    (k = 1.0000)
R4:  X1  →  D1    (k = 1.0000)
R5:  ∅  →  B1    (k = 6.0000)
R6:  B1  →  ∅    (k = 2.0000)
R7:  D1  →  B1    (k = 0.1000)
R8:  D1  →  ∅    (k = 0.0500)
R9:  ∅  →  X2    (k = 1.0000)
R10:  2X2 + Y2  →  3X2    (k = 1.0000)
R11:  B2 + X2  →  D2 + Y2    (k = 1.0000)
R12:  X2  →  D2    (k = 1.0000)
R13:  ∅  →  B2    (k = 5.0000)
R14:  B2  →  ∅    (k = 2.0000)
R15:  D2  →  B2    (k = 0.1000)
R16:  D2  →  ∅    (k = 0.0500)
R17:  ∅  →  X3    (k = 0.8000)
R18:  2X3 + Y3  →  3X3    (k = 1.0000)
R19:  B3 + X3  →  D3 + Y3    (k = 1.0000)
R20:  X3  →  D3    (k = 1.0000)
R21:  ∅  →  B3    (k = 5.0000)
R22:  B3  →  ∅    (k = 2.0000)
R23:  D3  →  B3    (k = 0.1000)
R24:  D3  →  ∅    (k = 0.0500)
R25:  ∅  →  X4    (k = 1.2000)
R26:  2X4 + Y4  →  3X4    (k = 1.0000)
R27:  B4 + X4  →  D4 + Y4    (k = 1.0000)
R28:  X4  →  D4    (k = 1.0000)
R29:  ∅  →  B4    (k = 7.0000)
R30:  B4  →  ∅    (k = 2.0000)
R31:  D4  →  B4    (k = 0.1000)
R32:  D4  →  ∅    (k = 0.0500)
R33:  ∅  →  X5    (k = 0.6000)
R34:  2X5 + Y5  →  3X5    (k = 1.0000)
R35:  B5 + X5  →  D5 + Y5    (k = 1.0000)
R36:  X5  →  D5    (k = 1.0000)
R37:  ∅  →  B5    (k = 4.0000)
R38:  B5  →  ∅    (k = 2.0000)
R39:  D5  →  B5    (k = 0.1000)
R40:  D5  →  ∅    (k = 0.0500)
R41:  X2 + Y1  →  X1 + Y2    (k = 0.0237)
R42:  X3 + Y2  →  X2 + Y3    (k = 0.0295)
R43:  X4 + Y3  →  X3 + Y4    (k = 0.0273)
R44:  X5 + Y4  →  X4 + Y5    (k = 0.0260)
R45:  X1 + Y5  →  X5 + Y1    (k = 0.0216)
R46:  X1 + Y3  →  X3 + Y1    (k = 0.0108)
R47:  X2 + Y4  →  X4 + Y2    (k = 0.0103)
R48:  X3 + Y5  →  X5 + Y3    (k = 0.0143)
R49:  X4 + Y1  →  X1 + Y4    (k = 0.0130)
R50:  X5 + Y2  →  X2 + Y5    (k = 0.0135)

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Input Species: B1, B2, B3, B4, B5
Output Species: X1, Y1, B1, D1, X2, Y2, B2, D2, X3, Y3, B3, D3, X4, Y4, B4, D4, X5, Y5, B5, D5
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